

Linear transformations

Definition

We say that a transformation $T : \underline{R}^n \rightarrow \underline{R}^m$ is a linear transformation if

- for all $u, v \in R^n$, $T(u + v) = T(u) + T(v)$, and
- for all scalars λ and $u \in R^n$, $T(\lambda u) = \lambda T(u)$.

Fact: If T is a linear transformation then $T(0) = 0$ and for all scalars c, d and vectors u, v , $T(cu + dv) = cT(u) + dT(v)$.

$$T(0) = T(\underset{\substack{\uparrow \\ \text{scalar}}}{0} \cdot \underset{\substack{\uparrow \\ \text{vector}}}{0}) = 0 T(0) = 0$$

If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear and $c, d \in \mathbb{R}$
and $u, v \in \mathbb{R}^n$

$$\begin{aligned} T(cu + dv) &= T(cu) + T(dv) \\ &= cT(u) + dT(v) \end{aligned}$$

In fact if $c_1, \dots, c_k \in \mathbb{R}$ and $u_1, \dots, u_k \in \mathbb{R}^n$
then if T is linear then

$$\begin{aligned} T(\underline{c_1 u_1 + \dots + c_k u_k}) &= c_1 \underline{T(u_1)} + c_2 \underline{T(u_2)} \\ &\quad + \dots + c_k \underline{T(u_k)}. \end{aligned}$$

If T is linear and $u_1, \dots, u_n \in \mathbb{R}^n$
($T: \mathbb{R}^n \rightarrow \mathbb{R}^m$) then

T is determined on the span of
 $u_1, \dots, u_n$ by $T(u_1), \dots, T(u_n)$.

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad T \text{ is linear and}$$
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \dots e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

then T is determined by $T(e_1) \dots T(e_n)$.

Why? Suppose $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ and we

want to compute $T \begin{pmatrix} x_1 \\ \vdots \\ \underline{x_n} \end{pmatrix}$

$$\underline{T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n)} = \underline{x_1 T(e_1)} + \underline{x_2 T(e_2)} \\ + \dots + \underline{x_n T(e_n)}.$$

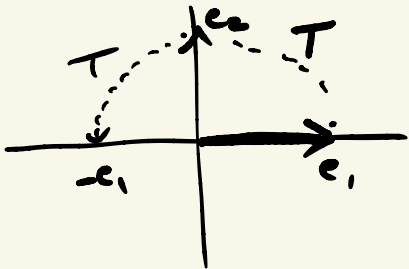
$$\text{So } T \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \underbrace{\begin{pmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{pmatrix}}_{\substack{\text{Standard matrix} \\ \text{for } T.}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Ex: Recall if A was $m \times n$ then

$T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T_A(x) = Ax$
matrix trans.

Ex: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, linear with

$$\underline{T(e_1) = e_2}, \quad \underline{T(e_2) = -e_1}$$



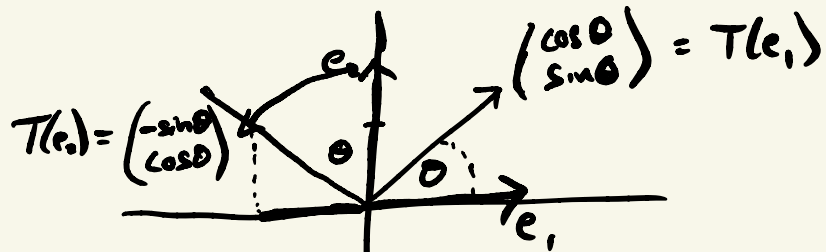
The standard
matrix for this
trans. is.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

Rotation by 90° . counterclockwise

Ex: What does rotation counterclockwise by the angle θ look like a matrix transformation?

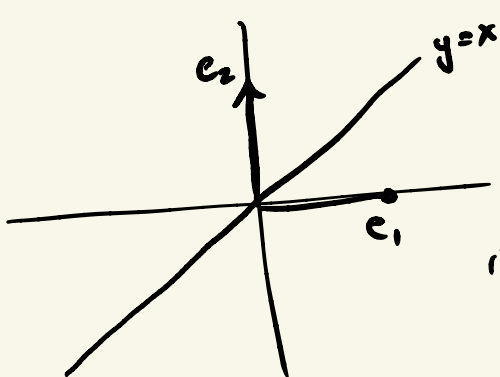


rotation matrix
by θ counter
clockwise. $\rightarrow$ So the standard matrix is

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Ex: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ reflection in the line $y=x$.

What is the standard matrix for T ?



$$T(e_1) = e_2$$

$$T(e_2) = e_1$$

The standard matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Standard matrix of a linear transformation

Theorem

Suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation then there is a unique $m \times n$ matrix A such that $T(x) = Ax$ for all x . In fact, if e_i is the vector with 1 in the i^{th} coordinate and 0 elsewhere for $i = 1, \dots, n$ then

$$A = [T(e_1) \quad T(e_2) \quad \dots \quad T(e_n)] .$$

This matrix A is called the standard matrix for T .

One-to-one and onto

Definition

We say that a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is

- **one-to-one** if whenever $u \neq v \in \mathbb{R}^n$ then $T(u) \neq T(v)$, and
- **onto** if the range of T is $\mathbb{R}^m$ i.e. for every $b \in \mathbb{R}^m$, there is $x \in \mathbb{R}^n$ such that $T(x) = b$.

Fact: If T is a linear transformation then the following are equivalent:

- for all $u \in \mathbb{R}^n$, if $T(u) = 0$ then $u = 0$.
- T is one-to-one.

Ex: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is linear given by.

$$T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Is T one-to-one? What values of $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

No!

satisfy $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$?

$$\begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

homogeneous
the matrix is
 2×3 so there
are inf. many solⁿ.

If $T(u) = T(v)$ where T is linear

then $T(u-v) = 0$. If $\text{ker } T = \{0\}$

we get $u=v$.

One-to-one and onto in terms of the standard matrix

Theorem

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, and A is the standard matrix for T . Then

- T is one-to-one iff the columns of A are linearly independent, and*
- T is onto $\mathbb{R}^m$ iff the span of the columns of A .*